

Generation Interconnection System Impact Study Report

For Queue Project Q22-2111

Whillock 138 kV

August 03, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2111. The Interconnection Customer proposes a 118.5 MW battery energy storage facility interconnecting at the Whillock 138 kV substation in the Pennsylvania area. The request was received on June 11, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 442 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,633,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Keswick-Walmart 138 KV Line
Contingency: Medora Substation-P.T.S Substation 345 KV
Loading: 142.8% (Rating: 242 MVA)
DFAX: 33.7%
Pre-project loading: 126.3%
2. Facility: Keswick-Arsenal Hill 138 KV
Contingency: Medora-P.T.S 345 KV
Loading: 138.0% (Rating: 268 MVA)
DFAX: 33.7%
Pre-project loading: 123.1%
3. Facility: Arsenal Hill Substation-Forbing Road 138 KV Line
Contingency: P.T.S-Medora 345 KV
Loading: 134.9% (Rating: 288 MVA)
DFAX: 33.7%
Pre-project loading: 121.0%
4. Facility: Forbing Rd-Whillock 138 KV
Contingency: P.T.S Substation-Medora 345 KV L
Loading: 132.0% (Rating: 309 MVA)
DFAX: 33.7%
Pre-project loading: 119.1%
5. Facility: Tecuya-Whillock 138 KV L
Contingency: Boyntonville-Whillock 138 KV Line
Loading: 131.3% (Rating: 201 MVA)
DFAX: 50.8%
Pre-project loading: 101.3%
6. Facility: Las Cruces 3602-E Vicksburg 138 KV L
Contingency: Boyntonville-Whillock 138 KV
Loading: 127.2% (Rating: 235 MVA)

DFAX: 50.8%

Pre-project loading: 101.6%

7. Facility: Tecuya-East Vicksburg 138 KV

Contingency: Boyntonville-Whillock 138 KV

Loading: 127.2% (Rating: 223 MVA)

DFAX: 50.8%

Pre-project loading: 100.2%

8. Facility: Las Cruces 302-Cadillac Renewable Energy 138 KV L

Contingency: Boyntonville-Whillock Substation 138 KV

Loading: 127.0% (Rating: 250 MVA)

DFAX: 50.8%

Pre-project loading: 102.9%

9. Facility: Lime Crk-Artesia 138 KV L

Contingency: Cadillac Renewable Energy-Hampton Rd 138 KV

Loading: 120.6% (Rating: 264 MVA)

DFAX: 22.2%

Pre-project loading: 110.7%

10. Facility: Pulliam Sw Yd-Southside Generating Sta 138 KV

Contingency: Forbing Road-Whillock 138 KV

Loading: 118.1% (Rating: 265 MVA)

DFAX: 9.1%

Pre-project loading: 114.1%

11. Facility: Whillock-Boyntonville 138 KV Line

Contingency: Cadillac Renewable Energy-Las Cruces 3602 138 KV Line

Loading: 116.9% (Rating: 343 MVA)

DFAX: 63.7%

Pre-project loading: 94.8%

12. Facility: Artesia-Boyntonville 138 KV

Contingency: Hampton Rd-Cadillac Renewable Energy 138 KV

Loading: 116.8% (Rating: 260 MVA)

DFAX: 22.2%

Pre-project loading: 106.6%

13. Facility: Hampton Rd-Cadillac Renewable Energy 138 KV Line

Contingency: Artesia-Lime Creek 138 KV

Loading: 116.7% (Rating: 276 MVA)

DFAX: 20.4%

Pre-project loading: 108.0%

14. Facility: Shepherds Central-Pulliam Sw Yd 138 KV Line

Contingency: Whillock-Forbing Road 138 KV

Loading: 115.7% (Rating: 260 MVA)

DFAX: 9.1%

Pre-project loading: 111.6%

15. Facility: Tuco-John Day Substation 138 KV Line

Contingency: Simpson Stuart Tap-Kempster 138 KV Line

Loading: 115.7% (Rating: 389 MVA)

DFAX: 5.5%

Pre-project loading: 114.1%

16. Facility: Shepherds Central Substation-Woodville 138 KV L

Contingency: Whillock-Forbing Road 138 KV line

Loading: 112.3% (Rating: 257 MVA)

DFAX: 9.1%

Pre-project loading: 108.1%

17. Facility: John Day-Matanzas 138 KV

Contingency: Simpson Stuart Tap Substation-Kempster 138 KV

Loading: 111.1% (Rating: 396 MVA)

DFAX: 5.5%

Pre-project loading: 109.5%

18. Facility: Kempster-Simpson Stuart Tap 138 KV

Contingency: John Day-Tuco 138 KV

Loading: 110.9% (Rating: 499 MVA)

DFAX: 6.5%

Pre-project loading: 109.4%

19. Facility: Filer 500-345 kV XFMR

Contingency: Tidd-Lerdo 500 KV line

Loading: 110.0% (Rating: 1,238 MVA)

DFAX: 8.0%

Pre-project loading: 109.2%

20. Facility: Richardson Woodhaven Substation-Fishkill Plains Substation 138 KV

Contingency: Simpson Stuart Tap-Fishkill Plains 345 KV

Loading: 109.0% (Rating: 281 MVA)

DFAX: 10.7%

Pre-project loading: 104.5%

21. Facility: Monongahela L&D4; Hydroelectric Project-Smithville 138 KV

Contingency: Simpson Stuart Tap-Kempster 138 KV

Loading: 107.5% (Rating: 231 MVA)

DFAX: 5.2%

Pre-project loading: 104.8%

22. Facility: Hendeson-Monongahela L&D4; Hydroelectric Project 138 KV

Contingency: Simpson Stuart Tap-Kempster 138 KV

Loading: 107.3% (Rating: 242 MVA)

DFAX: 5.2%

Pre-project loading: 104.7%

23. Facility: Fishkill Plains 500/345 kV Auto

Contingency: Fishkill Plains-Dillard-Orange 345 KV

Loading: 106.9% (Rating: 869 MVA)

DFAX: 10.0%

Pre-project loading: 105.6%

24. Facility: Beaver Valley Substation-Aes Beaver Vly Partners Beaver Valley 345 KV

Contingency: AesBeaver Vly Partners Beaver Valley-Midland 345 KV

Loading: 105.0% (Rating: 1,200 MVA)

DFAX: 5.7%

Pre-project loading: 104.4%

25. Facility: Lewsville-Williams 138 KV

Contingency: Dos Amigos-Ontonagon Substation 138 KV Line

Loading: 103.4% (Rating: 326 MVA)

DFAX: 6.5%

Pre-project loading: 101.1%

26. Facility: Pinkerton-Firstenergy Mitchell Power Sta 138 KV line

Contingency: Simpson Stuart Tap 345 kV/138 kV TX

Loading: 103.2% (Rating: 229 MVA)

DFAX: 11.5%

Pre-project loading: 97.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Keswick-Walmart 138 KV Line	Rebuild line with higher-capacity conductor	\$2,147,000	31% (\$656,000)
Keswick-Arsenal Hill 138 KV	Rebuild line with higher-capacity conductor	\$2,315,000	31% (\$708,000)
Arsenal Hill Substation-Forbing Road 138 KV Line	Rebuild line with higher-capacity conductor	\$1,131,000	31% (\$346,000)
Forbing Rd-Whillock 138 KV	Rebuild line with higher-capacity conductor	\$2,278,000	31% (\$696,000)
Whillock-Tecuya 138 KV	Rebuild line with higher-capacity conductor	\$2,746,000	48% (\$1,327,000)
Las Cruces 3602-E Vicksburg 138 KV line	Rebuild line with higher-capacity conductor	\$595,000	48% (\$288,000)
Tecuya-East Vicsburg 138 KV	Rebuild line with higher-capacity conductor	\$1,400,000	48% (\$676,000)
Las Cruces 302-Cadillac Renewable Energy 138 KV L	Rebuild line with higher-capacity conductor	\$5,644,000	48% (\$2,726,000)
Lime Crk-Artesia 138 KV L	Rebuild line with higher-capacity conductor	\$1,485,000	18% (\$264,000)
Pulliam Sw Yd-Southside Generating Sta 138 KV	Rebuild line with higher-capacity conductor	\$1,255,000	10% (\$119,000)
Boyntonville-Whillock 138 KV	Rebuild line with higher-capacity conductor	\$1,405,000	46% (\$650,000)
Artesia-Boyntonville 138 KV	Rebuild line with higher-capacity conductor	\$3,392,000	18% (\$604,000)
Hampton Rd-Cadillac Renewable Energy 138 KV Line	Rebuild line with higher-capacity conductor	\$4,278,000	18% (\$785,000)
Shepherds Central-Pulliam Sw Yd 138 KV Line	Rebuild line with higher-capacity conductor	\$1,763,000	10% (\$167,000)
Tuco-John Day Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$1,625,000	11% (\$177,000)
Shepherds Central Substation-Woodville 138 KV L	Rebuild line with higher-capacity conductor	\$1,094,000	10% (\$104,000)
John Day-Matanzas 138 KV Line	Rebuild line with higher-capacity conductor	\$2,194,000	11% (\$238,000)
Kempster-Simpson Stuart Tap 138 KV	Rebuild line with higher-capacity conductor	\$3,962,000	17% (\$665,000)
Filer 500-345 kV XFMR	Replace transformer with higher-rated unit	\$30,023,000	4% (\$1,302,000)
Fishkill Plains-Richardson Woodhaven 138 KV	Reconductor line	\$1,587,000	32% (\$508,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Smithville-Monongahela L&D4; Hydroelectric Project 138 KV	Reconductor line	\$3,678,000	12% (\$449,000)
Hendeson-Monongahela L&D4; Hydroelectric Project 138 KV	Reconductor line	\$1,068,000	12% (\$130,000)
Fishkill Plains 500/345 kV Auto	Replace transformer with higher-rated unit	\$23,108,000	11% (\$2,501,000)
Beaver Valley-Aes Beaver Vly Partners Beaver Vly 345 KV	Reconductor line	\$4,390,000	4% (\$164,000)
Lewsville-Williams 138 KV	Reconductor line	\$391,000	13% (\$53,000)
Pinkerton-Firstenergy Mitchell Power Sta 138 KV	Reconductor line	\$1,490,000	22% (\$330,000)
	Total allocated to project		\$16,633,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.